

Partial Derivatives Worksheet

Find all first order partial derivatives.

1.) $f(x, y) = (xy - 1)^2$

2.) $f(x, y) = \frac{1}{x + y}$

3.) $f(x, y) = xe^{x - y^3}$

4.) $f(x, y) = x^y$

5.) $f(x, y) = x \ln(xy) + y^2$

6.) $f(x, y, z) = x - \sqrt{y^2 + z}$

Find all second order partial derivatives (you can use Clairaut's Theorem).

1.) $f(x, y) = x^2 y + \ln(y^2 - x)$

2.) $f(x, y) = e^{x^4 y^6} - \frac{y^3}{x}$

3.) $f(x, y, z) = \frac{x^2 y^6}{z^3} - 2x^6 z + 8y^{-3} x^4 + 4z^2$